



Cumulative incidence of second primary cancers in a large nationwide cohort of Danish cancer survivors: a population-based retrospective cohort study

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Summary

Background A new primary cancer is a serious late effect of a pre-existing cancer diagnosis, and can be attributed to hereditary cancer syndromes, immune or hormonal factors, cancer treatment, or modifiable lifestyle or environmental factors. We investigated the absolute and relative incidence of second primary cancers in a large cohort of Danish cancer survivors. Furthermore, we examined the association between alcohol-related, smoking-related, virus-related, and hormone-related first and second primary cancers.

Methods In this retrospective cohort study, we identified a cohort of Danish adults (aged ≥ 40 years) diagnosed with cancer from Jan 1, 1997, to Dec 31, 2014 and alive 1 year after diagnosis. Follow-up was from date of first cancer diagnosis and lasted up to 24 years, ending on Dec 31, 2020. Cohort identification and information on second primary cancers was obtained from the Danish Cancer Registry, and comorbidity and sociodemographic information was obtained from Danish population-based registries. Overall, and for 27 cancer types, cumulative incidence functions and Cox proportional hazard regression models were used to estimate the incidence of second primary cancer and death, and hazard ratios (HRs) and 95% CIs of second primary cancer adjusted for sex, age and year of diagnosis, cohabitation status, income, and comorbidity.

Findings 457 334 Danish adults were included in our study (230 150 [50.3%] male individuals and 227 184 [49.7%] female individuals; median age at diagnosis 68.3 years, IQR 59.7–76.6; median follow-up 3.6 years, IQR 0.6–9.3). The cumulative incidence of second primary cancer increased over time from 6.3% (95% CI 6.2–6.4) 5 years after diagnosis to 10.5% (10.4–10.6) 10 years after diagnosis and to 13.5% (13.4–13.7) 15 years after diagnosis. The highest cumulative incidence of second primary cancer 10 years after diagnosis was observed in survivors of cancers in the larynx (21.8%, 20.5–23.1), oropharynx and oral cavity (19.5%, 18.7–20.3), and bladder and urinary tract (18.5%, 18.0–19.0). Survivors of cancers related to alcohol (HR 1.09, 95% CI 1.06–1.13), smoking (1.73, 1.68–1.78), diet high in red or processed meat (1.32, 1.24–1.39), or virus (1.23, 1.13–1.35) were at increased risk of developing a second cancer with the same aetiology, whereas having had a hormone-related first cancer was associated with lower risk of a second hormone-related cancer (0.77, 0.73–0.81).

Interpretation Our results could help optimise prevention efforts targeting modifiable risk factors to reduce risk of developing a second primary cancer.

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Introduction

Across almost all cancer sites, survival is increasing, leaving time for patients to become long-term survivors. This leaves them at risk of developing late effects following cancer treatment, with potential impact on their quality of life and daily life. New primary cancer is a serious late effect that occurs in 2–37% of patients with cancer during their lifetime, depending on age, first primary cancer type, follow-up time, and definitions used.^{1–5} New primary cancer can be attributed to several factors including genetic susceptibility, immune or hormonal factors, previous cancer treatment, and potentially modifiable risk factors such as lifestyle and environmental factors.⁶ The shared risk factors linked to

the exposures that led to the first cancer, such as smoking or excessive alcohol use, can contribute to development of a new primary cancer in the same or other organs.⁷ However, the association between modifiable shared risk factors and risk of new primary cancer among people who previously had cancer remains understudied and often the aetiology is complex, with combinations of shared risk factors.⁸

Previous epidemiological reports investigating the risk of new primary cancer have primarily focused on risk in childhood or adolescent cancer cohorts,^{9,10} included specific cancers or small cohorts, or had short follow-up.¹¹ One of the few recent large studies among adults compared data from 1.54 million individuals who were

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Research in context

Evidence before this study

We did a systematic search of PubMed, Embase, CINAHL, PsychInfo, and the Cochrane Library for articles published in English between Jan 1, 2000, and Feb 27, 2017. Subsequent searches were done in June 15, 2021, and Dec 1, 2022. The search terms were “new primary cancer”, “secondary primary cancer”, “subsequent primary cancer”, “second primary malignancies”, “second malignant neoplasms”, “late effects”, “late toxicity”, “long term effects”, “chemotherapy late effects” “radiation late effects”, “cancer survivor”, “survivorship”, “post treatment”, “post therapy”, “consequential effect”, “neoplasm”, and “adult cancer”. We searched the titles and abstracts of published articles to identify relevant studies. Studies were considered if they assessed new primary cancer and other late effects, long-term symptoms, or symptom burden in adult cancer survivors (age ≥ 40 years at the time of diagnosis). Although the search revealed a very large number of studies ($>18\,000$), the studies on new primary cancer in adult cancer survivors are fragmented and unequally distributed by cancer type. In general, the previous studies were investigating childhood cancer survivors, or were based on smaller cohorts or specific cancers. Very few studies provided an overview of new primary cancers in adult cancer survivors in general, and these studies are limited by small numbers of study participants, self-reported data, findings collapsed for cancer types, and disregard of comorbidity before the cancer diagnosis and the socioeconomic positions of study participants. In a previous study, including more than 458 000 Danish survivors and a

comparison population of more than 2.1 million cancer-free individuals, we showed that, overall, cancer survivors are at higher risk of hospitalisation for a broad range of incident diseases after cancer diagnosis than the cancer-free comparison population. However, slightly higher risks of new primary cancer were seen in only a few of the investigated cancer types whereas for the rest of the cancer types the risk of new primary cancer was equal or lower than that in the general population. Although our systematic search showed there are abundant publications on new primary cancers in cancer survivors, many previous studies used relative measures to examine the risk of new primary cancer compared with those expected in the general population and only few examine the cumulative incidence measures that might be more clinically relevant and of individual importance to the survivors and their families.

Added value of this study

We present an updated overview of risk of second primary cancer based on a recent national cohort of more than 457 000 survivors with long follow-up. We estimate cumulative incidence rates, accounting for risk of death, and investigate how modifiable lifestyle and other environmental factors might affect the pattern of new primary cancers.

Implications of all the available evidence

Closer monitoring for new primary cancer among high-risk cancer survivors might result in earlier detection and management of this serious late effect in cancer survivors.

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registered in 12 Surveillance, Epidemiology and End Results (SEER) registries from 1992 to 2011 with a first primary cancer with a reference population, and reported standardised incidence ratios of 1.10 (95% CI 1.09–1.11) among female individuals and 1.11 (1.10–1.12) among male individuals (excluding those with prostate cancer), with increased risk of new primary cancer found for 21 of 31 investigated cancer types in female individuals and 18 of 30 types in male individuals.¹² However, such previous studies primarily used relative measures to compare the risk of a second primary cancer among patients with cancer with the risk of a novel primary cancer in the general population. Although relative measures can provide important information on excess risk for a given patient population, cumulative incidence measures might improve clinical relevance and be important to survivors and their families. Furthermore, the high mortality among some groups of patients with cancer (eg, lung cancer, pancreatic cancer, or brain tumours) makes it difficult to assess the true risk of new primary cancer in cancer survivors when compared with the risk of a novel primary cancer in the cancer-free general population. Curtis and colleagues⁵ examined the risk of new malignancies among more than 2 million cancer survivors of all ages using data from nine SEER

Programs registered in the period 1973–2000, reporting cumulative incidence rates of new primary cancer of 5.0% at 5 years and 8.4% at 10 years after initial cancer diagnosis. Apart from this work, only limited updated knowledge exists of the cumulative incidence of new primary cancer across adult-onset cancer types and time.

We investigated the absolute and relative incidence of new (ie, second) primary cancer in a nationwide cohort of Danish adult cancer survivors (defined as alive 1 year after diagnosis) using Danish population-based nationwide registries. We also evaluated how common aetiological exposures for a first cancer were associated with development of a second primary cancer, by investigating the association between alcohol-related, smoking-related, virus-related, and hormone-related first and second cancers.

Methods

Study population and data sources

The cohort has been described previously,¹³ and a study synopsis is provided in the appendix (pp 12–19). In brief, this study is based on updated nationwide registry data for 457 334 Danish adult cancer survivors who were diagnosed with their first primary cancer between Jan 1, 1997, and Dec 31, 2014. All cohort members were

See Online for appendix

	Total (n=457 334)	Male individuals (n=230 150)	Female individuals (n=227 184)
Age, years			
40–49	35 781 (7.8%)	12 612 (5.5%)	23 169 (10.2%)
50–59	81 198 (17.8%)	36 211 (15.7%)	44 987 (19.8%)
60–69	136 381 (29.8%)	73 600 (32.0%)	62 781 (27.6%)
70–79	129 948 (28.4%)	73 376 (31.9%)	56 572 (24.9%)
≥80	74 026 (16.2%)	34 351 (14.9%)	39 675 (17.5%)
Median (IQR)	68.3 (59.7–76.6)	69.1 (61.5–76.4)	67.3 (57.9–76.8)
Year of diagnosis			
1997–2000	86 935 (19.0%)	41 706 (18.1%)	45 229 (19.9%)
2001–04	93 227 (20.4%)	45 902 (19.9%)	47 325 (20.8%)
2005–07	77 144 (16.9%)	39 661 (17.2%)	37 483 (16.5%)
2008–10	84 687 (18.5%)	42 911 (18.6%)	41 776 (18.4%)
2011–14	115 341 (25.2%)	59 970 (26.1%)	55 371 (24.4%)
Cohabitation status			
Living with partner	274 189 (60.0%)	156 626 (68.1%)	117 563 (51.8%)
Living alone	182 809 (40.0%)	73 309 (31.9%)	109 500 (48.2%)
Missing	336 (<0.1%)	215 (<0.1%)	121 (<0.1%)
Charlson Comorbidity Index*			
0	382 077 (83.5%)	186 751 (81.1%)	195 326 (86.0%)
1	56 125 (12.3%)	31 223 (13.6%)	24 902 (11.0%)
≥2	19 132 (4.2%)	12 176 (5.3%)	6956 (3.1%)
Education†			
Short	139 845 (30.6%)	65 669 (28.5%)	74 176 (32.7%)
Medium	194 522 (42.5%)	104 272 (45.3%)	90 250 (39.7%)
Higher	76 692 (16.8%)	38 860 (16.9%)	37 832 (16.7%)
Missing	46 275 (10.1%)	21 349 (9.4%)	24 926 (11.0%)
Income quintile‡			
1	93 689 (20.5%)	48 302 (21.0%)	45 387 (20.0%)
2	95 912 (21.0%)	48 259 (21.0%)	47 653 (21.0%)
3	90 175 (19.7%)	45 152 (19.6%)	45 023 (19.8%)
4	87 513 (19.1%)	43 576 (18.9%)	43 937 (19.3%)
5	89 899 (19.7%)	44 781 (19.5%)	45 118 (19.9%)
Missing	146 (<0.1%)	80 (<0.1%)	66 (<0.1%)
First primary cancer type			
Oropharynx and oral cavity	9342 (2.0%)	6375 (2.8%)	2967 (1.3%)
Oesophagus	6402 (1.4%)	4697 (2.0%)	1705 (0.8%)
Stomach	8245 (1.8%)	5368 (2.3%)	2877 (1.3%)
Small intestine	1278 (0.3%)	655 (0.3%)	623 (0.3%)
Colon	38 814 (8.5%)	18 823 (8.2%)	19 991 (8.8%)
Rectum	22 166 (4.9%)	13 318 (5.8%)	8848 (3.9%)
Anus	1616 (0.4%)	496 (0.2%)	1120 (0.5%)
Liver	4786 (1.1%)	3247 (1.4%)	1539 (0.7%)
Pancreas	13 069 (2.9%)	6488 (2.8%)	6581 (2.9%)
Larynx	3950 (0.9%)	3254 (1.4%)	696 (0.3%)
Lung	61 456 (13.4%)	33 227 (14.4%)	28 229 (12.4%)
Sarcoma	4284 (0.9%)	2620 (1.1%)	1664 (0.7%)
Malignant melanoma	19 535 (4.3%)	9413 (4.1%)	10 122 (4.5%)
Breast	67 991 (14.9%)	419 (0.2%)	67 572 (29.7%)
Cervix	4416 (1.0%)	0	4416 (1.9%)
Endometrium	11 173 (2.4%)	0	11 173 (4.9%)
Ovary	8660 (1.9%)	0	8660 (3.8%)

(Table 1 continues on next page)

aged 40 years or older at the time of diagnosis and were diagnosed 1 year before inclusion. All patients with cancer were identified in the nationwide Danish Cancer Registry, which contains information on type of cancer and date of diagnosis on all Danes with a diagnosis of cancer since 1943.¹⁴

By the use of the Danish unique personal identification number (PIN) assigned via the Danish Civil Registration system to all Danish citizens at birth since 1968,¹⁵ each person in the cohort was linked to the files of the National Patient Register¹⁶ to attain information on medically verified somatic diseases up to Dec 31, 2018. Since 1978, the National Patient Register holds information on diagnoses coded according to the International Classification of Diseases 8th revision (ICD-8; from 1978 to 1993) and 10th revision (ICD-10; from 1994 onwards) for all hospitalisations for somatic diseases and, since 1995, also outpatient and emergency visits.¹⁶ We obtained information on somatic conditions from 3 years before and up to the diagnosis of the first cancer and calculated Charlson Comorbidity Index (CCI)¹⁷ scores to provide an overall score for comorbidity based on 19 selected conditions, excluding the cancers in question. The CCI score was grouped as 0, 1, or 2 or higher, with a score of 0 indicating no comorbidity, a score of 1 indicating mild comorbidity, and a score of 2 or higher indicating moderate to severe comorbidity.

The cohort was further linked to the Danish nationwide administrative social registries, updated annually by Statistics Denmark since 1981,¹⁸ to obtain socio-demographic information on highest attained education (short, medium, and higher), income (quintiles), and cohabitation status (living with partner or living alone) at the time of diagnosis. Information on sex refers to biological sex and was based on the last digit of the participant's PIN (male individuals have an odd number and female individuals an even number).

There was no contact with cohort members. According to Danish legislation, informed consent or ethical board review are not required in a register-based study with no contact with individuals. We adhered to the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines.

Information on second primary cancer was obtained from the Danish Cancer Registry.¹⁴ The definition of multiple primaries is described in the appendix (p 11) and the included first and second primary cancers sites are shown in the appendix (p 1). To assure a general and clinically useful representation of cancer sites, we selected the 27 most common first primary cancer types (except for non-melanoma skin cancer) that fulfilled the inclusion criteria for the study. Individuals diagnosed with other cancers were grouped together into a single category, but this category was included only in estimations of the cumulative incidence of second primary cancer overall, not in the estimations for each individual type of second primary cancer.

To investigate whether shared aetiological risk factors for the first cancer contribute to the risk of a second primary cancer, we grouped the survivors into five groups by major aetiology of the first cancer. Only cancers with aetiological risk factors that were classified as group 1 (carcinogenic to humans) or group 2A (probably carcinogenic to humans) by the International Agency for Research in Cancer (IARC), or otherwise considered as causally related to cancer with epidemiological evidence, were included. The groups included cancers related to alcohol,¹⁹ smoking,¹⁹ a diet high in red and processed meat,²⁰ viruses,²¹ and hormones.²² ICD-10 codes are provided in the appendix (p 2).

Statistical analysis

A diagnosis of a second primary cancer was the primary outcome of the study. All survivors were followed up for a second primary cancer from the date of the first cancer diagnosis until the date of event or death (competing event), censoring at emigration, or end of follow-up (Dec 31, 2020), whichever came first. Only the first date of diagnosis for a second primary cancer was counted, to ensure it was the second primary cancer. The main reason for censoring was reaching the end of the study period and was thus non-informative.

To account for death as a competing event, we used cumulative incidence functions to estimate the risk of second primary cancer for different groups of cancer survivors. Multivariable Cox proportional hazard regression models were used to estimate the cause-specific hazard ratios (HRs) with the corresponding 95% CIs for a second primary cancer among the ten first primary cancers with the highest cumulative incidence of any second primary cancer after 10 years. Time since the first primary cancer diagnosis was used as the underlying time scale. We used a causal directed acyclic graph to identify possible confounders and mediators on the association between the first primary cancer and a second primary cancer. We identified sex, age and year of diagnosis, cohabitation status, income, and comorbidity as confounders, and adjusted for these in the multivariable analyses. Survivors with missing data for cohabitation status and income were excluded in these analyses.

Shared aetiological risk factors between first and second primary cancer were also investigated using multivariable Cox proportional hazard regression models using the same confounders. Because some of the cancers in some of the aetiological groups overlapped, analyses included three possible events as competing events: development of a second primary cancer in the aetiological group of interest; death; and development of second primary cancer in one of other aetiological groups.

The assumptions underlying the Cox proportional model were assessed using plots of Schoenfeld residuals. When the proportional hazards assumption was not

fulfilled for potential confounders, the variable was included in the baseline hazard. To describe the effect of first primary cancer type, the follow-up time was split into time bands; time band-specific HRs were estimated,

	Total (n=457 334)	Male individuals (n=230 150)	Female individuals (n=227 184)
(Continued from previous page)			
Prostate	53 744 (11.8%)	53 744 (23.4%)	0
Testis	1833 (0.4%)	1833 (0.8%)	0
Kidney	9171 (2.0%)	5848 (2.5%)	3323 (1.5%)
Bladder and urinary tract	24 226 (5.3%)	18 021 (7.8%)	6205 (2.7%)
Brain and nervous system	14 870 (3.3%)	6738 (2.9%)	8132 (3.6%)
Thyroid	2283 (0.5%)	668 (0.3%)	1615 (0.7%)
Hodgkin lymphoma	687 (0.2%)	417 (0.2%)	270 (0.1%)
Non-Hodgkin lymphoma	8847 (1.9%)	4887 (2.1%)	3960 (1.7%)
Multiple myeloma	5114 (1.1%)	2832 (1.2%)	2282 (1.0%)
Leukaemia	11 324 (2.5%)	6558 (2.9%)	4766 (2.1%)
Other	38 052 (8.3%)	20 204 (8.8%)	17 848 (7.9%)

Data are n (%) unless otherwise specified. Sex refers to biological sex and is based on the last digit of the participant's Danish personal identification number assigned at birth (male individuals are given an odd number and female individuals an even number). In Denmark, a person can legally change gender and receive a new number reflecting this change, but the current system only allows categorisation as either male or female. *Comorbid somatic conditions were grouped using the Charlson Comorbidity Index, according to accumulated sum of scores. This index provides an overall score for comorbidity based on a composite of values weighted by level of severity assigned to 19 selected conditions scored from 1 to 6. For the purposes of this study, no cancers were included in the index. Scores are summarised on the basis of information on hospitalisations from 3 years before diagnosis and onwards. The weights are assigned for each condition that a patient has, and the total is the score. For example, chronic pulmonary disease (1) and hemiplegia (2) give a total score of 3. Score 1: myocardial infarct, congestive heart failure, peripheral vascular disease, cerebrovascular disease, dementia, chronic pulmonary disease, connective tissue disease, ulcer disease, mild liver disease, diabetes types 1 and 2. Score 2: hemiplegia, moderate-to-severe renal disease, diabetes with end-organ damage type 1 or type 2. Score 3: moderate-to-severe liver disease. Score 6: AIDS. †Short education: mandatory education of up to 7 years and 9 years for patients born before and after Jan 1, 1958, respectively; medium education: highest grade of primary school, secondary school, or vocational education (8–12 years and 10–12 years for patients born before or after Jan 1, 1958, respectively); higher education: >12 years of education. ‡Household income after taxation and interest adjusted for number of people in the household; categorised on the basis of sex-specific distribution of household disposable income per person.

Table 1: Descriptive statistics at time of diagnosis of 457 334 adult Danish cancer survivors, overall and by sex

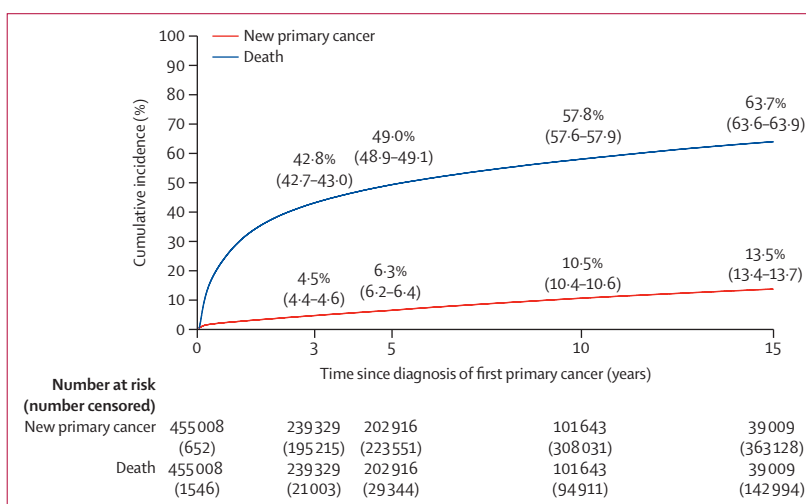


Figure 1: Cumulative incidence of second primary cancer and death after any first primary cancer. The cumulative incidence estimates are shown with 95% CIs in parentheses.

	5-year cumulative incidence	10-year cumulative incidence
Total	6.3% (6.2–6.4)	10.5% (10.4–10.6)
Sex		
Male	7.2% (7.1–7.3)	11.5% (11.4–11.6)
Female	5.3% (5.3–5.4)	9.4% (9.3–9.5)
Age at diagnosis, years		
40–49	3.6% (3.4–3.8)	7.3% (7.0–7.6)
50–59	5.2% (5.1–5.4)	10.0% (9.8–10.2)
60–69	7.2% (7.0–7.3)	12.7% (12.5–12.9)
70–79	7.6% (7.4–7.7)	11.7% (11.5–11.8)
≥80	5.0% (4.8–5.1)	6.3% (6.2–6.5)
Cohabitation status		
Living with partner	6.6% (6.5–6.7)	11.3% (11.2–11.4)
Living alone	5.8% (5.7–5.9)	9.2% (9.0–9.3)
Charlson Comorbidity Index		
0	6.4% (6.3–6.5)	10.8% (10.7–10.9)
1	5.8% (5.6–6.0)	8.7% (8.5–8.9)
≥2	5.7% (5.4–6.1)	8.0% (7.6–8.4)
Education		
Higher	6.5% (6.3–6.7)	11.5% (11.3–11.7)
Medium	6.8% (6.6–6.9)	11.4% (11.3–11.6)
Short	6.2% (6.1–6.4)	10.1% (9.9–10.2)
Income quintile		
1	6.4% (6.3–6.6)	10.4% (10.2–10.6)
2	6.3% (6.2–6.5)	10.5% (10.3–10.7)
3	6.4% (6.2–6.5)	10.7% (10.5–10.9)
4	6.2% (6.0–6.3)	10.4% (10.2–10.6)
5	6.2% (6.0–6.4)	10.4% (10.2–10.6)

Data in parentheses are 95% CIs. Individuals with missing data on cohabitation status and education were excluded. All covariates are measured at time of diagnosis. For descriptions of Charlson Comorbidity Index, education, and income quintile, see table 1 legend.

Table 2: 5-year and 10-year cumulative incidence of second primary cancer in adult Danish cancer survivors, by age, sex, comorbidity, and socioeconomic variables at time of diagnosis

and a Wald test was used to test a time-invariant effect of first primary cancer over the follow-up time.

A significance level of 0.05 was used. Analyses were carried out using Stata release 14. Graphs were completed in R version 4.1.0.

Role of the funding source

The funders of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Results

We included 457 334 cancer survivors (230 150 male individuals and 227 184 female individuals), of whom 56 114 (12.3%) developed a second primary cancer during up to 24 years of follow-up (distributions shown in appendix p 3). The median follow-up time was 3.6 years (IQR 0.6–9.3, range 0.0–24.0). In all, 120 278 (26.3%) of the participants were censored, mainly due to end of follow-up (119 364 [26.1%]) and emigration (915 [0.2%]).

The ratio between male individuals and female individuals was 1:1 and the median age at time of diagnosis was 69.1 years (IQR 61.5–76.4) for male individuals and 67.3 years (57.9–76.8) for female individuals (table 1). We did not obtain data on race or ethnicity.

For all cancers combined, the cumulative incidence of a second primary cancer was 6.3% (95% CI 6.2–6.4) 5 years after diagnosis, 10.5% (10.4–10.6) 10 years after diagnosis, and 13.5% (13.4–13.7) 15 years after diagnosis (figure 1).

419 282 (91.7%) cancer survivors had one of the 27 most common first primary cancer types, and the remaining 38 052 (8.3%) had other cancer types. By site, the cumulative incidence rates for second primary cancers

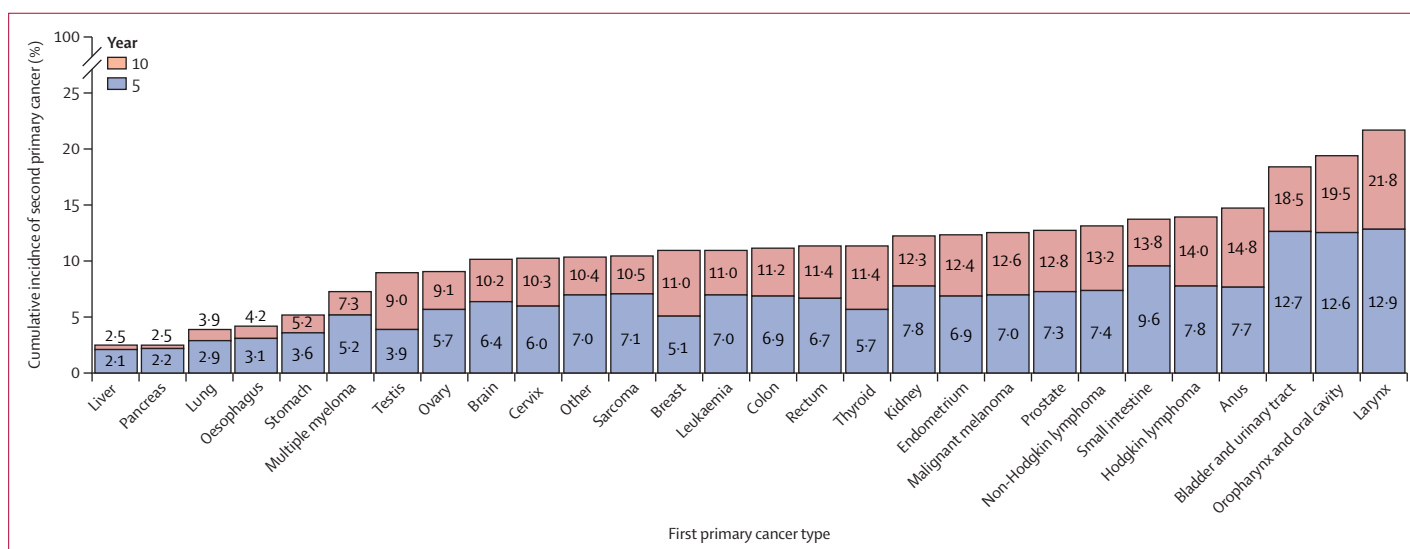
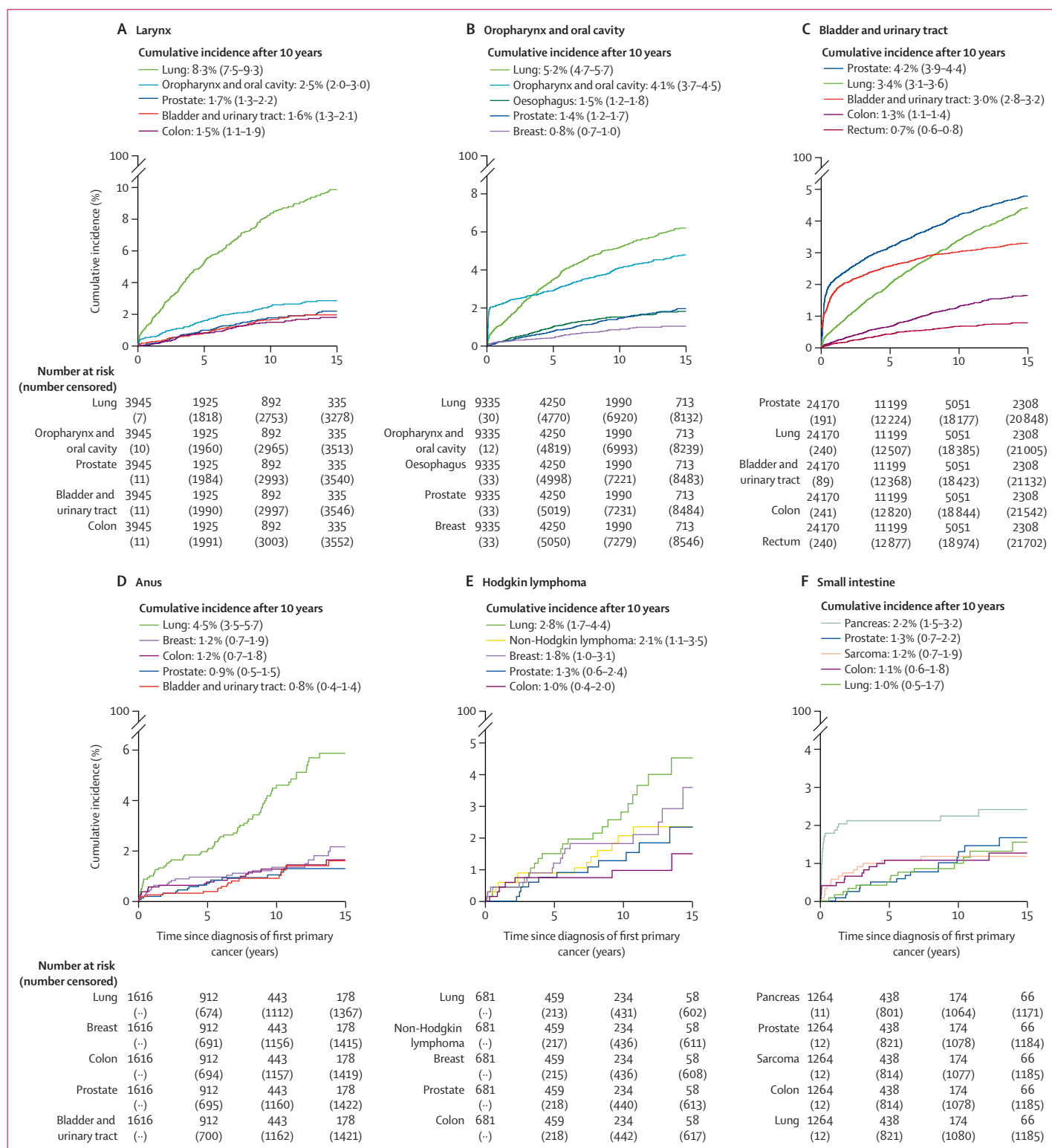


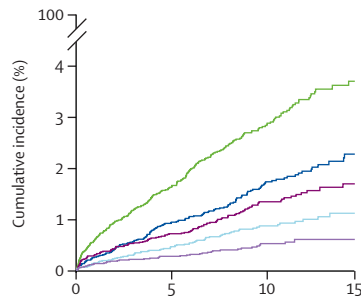
Figure 2: 5-year and 10-year cumulative incidence of any second primary cancer in adult (aged ≥40 years) cancer survivors of the 27 most common first primary cancer types
See appendix (p 4) for 95% CIs.



(Figure 3 continues on next page)

G Non-Hodgkin lymphoma**Cumulative incidence after 10 years**

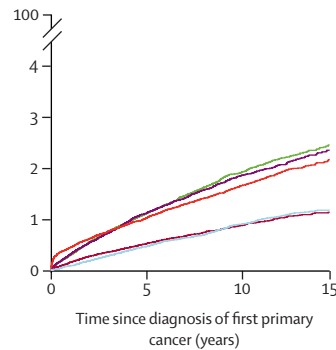
— Lung: 2.8% (2.4–3.2)
 — Prostate: 1.7% (1.4–2.0)
 — Colon: 1.3% (1.1–1.6)
 — Malignant melanoma: 0.9% (0.7–1.1)
 — Breast: 0.5% (0.4–0.7)

**Number at risk
(number censored)**

Lung	8800 (21)	4950 (3713)	2156 (6425)	381 (8170)
Prostate	8800 (24)	4950 (3773)	2156 (6521)	381 (8656)
Colon	8800 (24)	4950 (3790)	2156 (6546)	381 (8309)
Malignant melanoma	8800 (23)	4950 (3815)	2156 (6581)	381 (8347)
Breast	8800 (23)	4950 (3828)	2156 (6609)	381 (8379)

H Prostate**Cumulative incidence after 10 years**

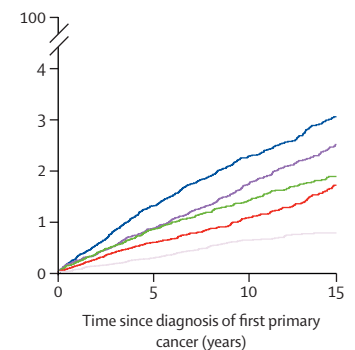
— Lung: 1.9% (1.8–2.0)
 — Colon: 1.8% (1.7–2.0)
 — Bladder and urinary tract: 1.6% (1.5–1.7)
 — Rectum: 0.9% (0.8–1.0)
 — Malignant melanoma: 0.9% (0.8–1.0)



Lung	53 605 (147)	32 506 (20 534)	13 292 (39 399)	2 588 (49 989)
Colon	53 605 (150)	32 506 (20 530)	13 292 (39 428)	2 588 (50 030)
Bladder and urinary tract	53 605 (114)	32 506 (20 578)	13 292 (39 521)	2 588 (50 115)
Rectum	53 605 (148)	32 506 (20 838)	13 292 (39 898)	2 588 (50 542)
Malignant melanoma	53 605 (151)	32 506 (20 866)	13 292 (39 894)	2 588 (50 536)

I Malignant melanoma**Cumulative incidence after 10 years**

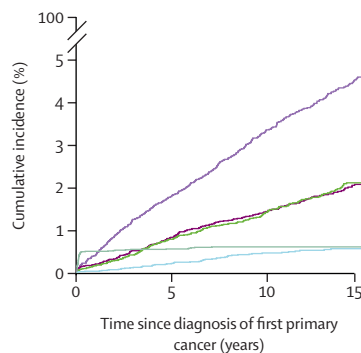
— Prostate: 2.2% (2.0–2.5)
 — Breast: 1.7% (1.5–1.9)
 — Lung: 1.4% (1.2–1.6)
 — Colon: 1.1% (0.9–1.2)
 — Brain: 0.6% (0.5–0.8)



Prostate	19 530 (13)	14 626 (4660)	7861 (11 177)	3417 (15 651)
Breast	19 530 (11)	14 626 (4744)	7861 (11 378)	3417 (15 753)
Lung	19 530 (13)	14 626 (4748)	7861 (11 427)	3417 (15 825)
Colon	19 530 (13)	14 626 (4796)	7861 (11 489)	3417 (15 879)
Brain	19 530 (12)	14 626 (4854)	7861 (11 563)	3417 (15 994)

J Endometrium**Cumulative incidence after 10 years**

— Breast: 3.3% (3.0–3.7)
 — Colon: 1.4% (1.2–1.7)
 — Lung: 1.4% (1.2–1.7)
 — Ovary: 0.6% (0.5–0.8)
 — Malignant melanoma: 0.5% (0.3–0.6)

**Number at risk
(number censored)**

Breast	11 159 (50)	7 635 (3328)	4 433 (6386)	2 083 (8662)
Colon	11 159 (50)	7 635 (3435)	4 433 (6580)	2 083 (8892)
Lung	11 159 (52)	7 635 (3440)	4 433 (6583)	2 083 (8892)
Ovary	11 159 (37)	7 635 (3464)	4 433 (6661)	2 083 (9012)
Malignant melanoma	11 159 (52)	7 635 (3502)	4 433 (6682)	2 083 (9026)

were generally lowest among survivors who had a first primary cancer with a high cumulative incidence of death, such as lung, pancreatic, or oesophageal cancer (appendix p 10).

Throughout the follow-up period, male individuals had higher overall cumulative incidence of second primary cancer than female individuals, and the highest rates of a second primary cancer were seen in people aged 60–69 years and 70–79 years at diagnosis (table 2). Survivors who were living with a partner at the time of diagnosis and those with higher or medium education had slightly higher cumulative incidence than did those who were living alone or had short education. Conversely, survivors who had comorbidity at the time of first cancer diagnosis (CCI score of 1 or ≥ 2) were generally less likely to be diagnosed with a second primary cancer during follow-up than were those without comorbidities (CCI score 0). Cumulative incidence was similar regardless of income quintile.

The highest cumulative incidence of a second primary cancer at 10 years after diagnosis of a first cancer was seen among individuals who had first primary cancer of the larynx (21·8%, 95% CI 20·5–23·1), oropharynx and oral cavity (19·5%, 18·7–20·3), and bladder and urinary tract (18·5%, 18·0–19·0). Most cancer types (17 of 27 investigated cancer types) had a cumulative incidence between approximately 10% and 15% (figure 2; appendix p 4). Survivors of liver, pancreas, and lung cancer had the lowest 10-year cumulative incidence of second primary cancer (liver: 2·5%, 2·1–3·0; pancreas: 2·5%, 2·3–2·8; and lung: 3·9%, 3·7–4·7). Lung cancer was the most frequent or second most frequent second primary cancer for seven of the ten first primary cancer types with the highest 10-year cumulative incidence of second primary cancer (figure 3). For the rest of the investigated cancer types, the site of the second primary cancer varied (appendix pp 5–6).

When calculating the HRs for a second cancer among survivors of the ten first primary cancers with the highest 10-year cumulative incidence of any second primary cancer, results showed that survivors after cancers in the larynx, oropharynx and oral cavity, bladder and urinary tract, anus, and small intestine had an increased hazard for developing a second primary cancer during follow-up, whereas the remaining five first primary cancer types

Figure 3: Cumulative incidence of second primary cancer in survivors after first primary cancer

Each panel shows a different first primary cancer site or type: larynx (A), oropharynx and oral cavity (B), bladder and urinary tract (C), anus (D), Hodgkin lymphoma (E), small intestine (F), non-Hodgkin lymphoma (G), prostate (H), malignant melanoma (I), and endometrium (J). For each first primary cancer, cumulative incidence rates of second primary cancers are shown, with 10-year cumulative incidence estimates and 95% CIs in parentheses. In some cases (year 0 in panels D and E), the numbers censored are not reported due to too few observations; Statistics Denmark does not allow data to be extracted from the registries if the total number of observations is less than 5.

	Overall	Risk factors for second primary cancer*				
		Alcohol	Smoking	Diet†	Virus	Hormone
Alcohol-related first primary cancer (n=152 350)						
Overall	1·01 (1·00–1·03)	1·09 (1·06–1·13)	1·17 (1·14–1·21)	1·13 (1·07–1·19)	0·95 (0·89–1·01)	0·93 (0·90–0·97)
Effect by time (years)						
0–2	0·91 (0·87–0·94)	1·43 (1·34–1·52)	1·12 (1·06–1·18)	1·36 (1·24–1·48)	1·07 (0·96–1·20)	0·79 (0·74–0·85)
2–5	1·01 (0·97–1·05)	1·00 (0·94–1·07)	1·20 (1·13–1·27)	1·15 (1·05–1·27)	0·93 (0·82–1·05)	0·98 (0·90–1·06)
5–10	1·12 (1·08–1·16)	1·04 (0·98–1·11)	1·23 (1·17–1·30)	1·10 (1·00–1·20)	0·91 (0·82–1·02)	1·05 (0·97–1·12)
≥10	1·02 (0·99–1·07)	0·89 (0·83–0·96)	1·15 (1·07–1·23)	0·88 (0·79–0·98)	0·89 (0·78–1·01)	0·95 (0·88–1·04)
Test for time-invariant effect‡	<0·0001	<0·0001	0·068	<0·0001	0·081	<0·0001
Smoking-related first primary cancer (n=145 346)						
Overall	1·34 (1·32–1·37)	1·46 (1·40–1·51)	1·73 (1·68–1·78)	0·99 (0·93–1·05)	1·25 (1·17–1·34)	1·40 (1·34–1·46)
Effect by time (years)						
0–2	1·35 (1·30–1·39)	1·37 (1·28–1·46)	1·79 (1·70–1·89)	0·73 (0·66–0·81)	1·30 (1·16–1·45)	1·47 (1·37–1·58)
2–5	1·41 (1·35–1·47)	1·47 (1·36–1·59)	1·77 (1·65–1·89)	1·06 (0·94–1·20)	1·25 (1·09–1·45)	1·45 (1·31–1·59)
5–10	1·34 (1·29–1·39)	1·55 (1·44–1·67)	1·76 (1·65–1·87)	1·18 (1·05–1·32)	1·26 (1·11–1·44)	1·34 (1·22–1·47)
≥10	1·26 (1·20–1·33)	1·49 (1·36–1·64)	1·49 (1·37–1·62)	1·24 (1·08–1·43)	1·14 (0·96–1·35)	1·27 (1·13–1·43)
Test for time-invariant effect‡	0·018	0·076	0·0019	<0·0001	0·66	0·12
Diet-related† first primary cancer (n=59 370)						
Overall	1·01 (0·98–1·03)	1·18 (1·13–1·23)	0·83 (0·79–0·86)	1·32 (1·24–1·39)	0·84 (0·77–0·91)	1·41 (1·35–1·48)
Effect by time (years)						
0–2	0·99 (0·95–1·04)	1·47 (1·36–1·58)	0·83 (0·77–0·90)	1·95 (1·77–2·15)	0·85 (0·73–1·00)	1·18 (1·08–1·30)
2–5	1·02 (0·97–1·07)	1·20 (1·10–1·31)	0·83 (0·76–0·91)	1·36 (1·21–1·52)	0·77 (0·65–0·92)	1·40 (1·26–1·55)
5–10	1·05 (1·01–1·10)	1·04 (0·95–1·13)	0·84 (0·77–0·91)	0·99 (0·88–1·12)	0·87 (0·75–1·02)	1·67 (1·53–1·83)
≥10	0·94 (0·89–1·00)	0·96 (0·86–1·07)	0·79 (0·71–0·88)	0·90 (0·77–1·06)	0·83 (0·69–1·01)	1·48 (1·32–1·66)
Test for time-invariant effect‡	0·026	<0·0001	0·89	<0·0001	0·75	<0·0001
Virus-related first primary cancer (n=43 611)						
Overall	1·10 (1·07–1·14)	1·14 (1·08–1·20)	1·19 (1·13–1·24)	1·02 (0·94–1·11)	1·23 (1·13–1·35)	1·19 (1·12–1·27)
Effect by time (years)						
0–2	1·02 (0·97–1·07)	1·05 (0·94–1·16)	1·13 (1·04–1·23)	0·95 (0·82–1·11)	1·44 (1·23–1·68)	0·97 (0·86–1·09)
2–5	1·16 (1·09–1·23)	1·13 (1·01–1·27)	1·14 (1·03–1·26)	1·04 (0·88–1·22)	1·20 (0·99–1·45)	1·35 (1·19–1·53)
5–10	1·12 (1·07–1·19)	1·16 (1·05–1·29)	1·30 (1·19–1·42)	1·08 (0·93–1·26)	1·31 (1·11–1·55)	1·24 (1·10–1·40)
≥10	1·15 (1·07–1·23)	1·24 (1·10–1·41)	1·19 (1·06–1·33)	1·03 (0·84–1·26)	0·83 (0·64–1·06)	1·37 (1·19–1·58)
Test for time-invariant effect‡	0·0048	0·19	0·11	0·72	0·0030	<0·0001
(Table 3 continues on next page)						

(Table 3 continues on next page)

(Continued from previous page)

Hormone-related first primary cancer (n=145 846)

Overall	0.74 (0.73–0.76)	0.75 (0.72–0.77)	0.71 (0.69–0.73)	0.96 (0.91–1.00)	0.86 (0.81–0.91)	0.77 (0.73–0.81)§
Effect by time (years)						
0–2	0.63 (0.61–0.65)	0.69 (0.65–0.74)	0.59 (0.56–0.63)	0.94 (0.86–1.02)	0.80 (0.72–0.90)	1.02 (0.93–1.12)
2–5	0.67 (0.65–0.70)	0.75 (0.70–0.80)	0.68 (0.64–0.72)	0.93 (0.85–1.02)	0.85 (0.76–0.96)	0.71 (0.64–0.78)
5–10	0.82 (0.79–0.84)	0.77 (0.72–0.81)	0.76 (0.72–0.80)	1.02 (0.94–1.11)	0.85 (0.77–0.94)	0.67 (0.61–0.73)
≥10	0.88 (0.84–0.91)	0.78 (0.73–0.84)	0.91 (0.85–0.97)	0.92 (0.83–1.03)	0.94 (0.83–1.07)	0.72 (0.65–0.80)
Test for time-invariant effect‡	<0.0001	0.063	<0.0001	0.33	0.31	<0.0001

Data are hazard ratio (95% CI) unless otherwise specified. Hazard ratios were adjusted for sex, age group, year group, cohabitation status, Charlson Comorbidity Index, and income quintile. Individuals with missing data on cohabitation status and income were excluded. All covariates are measured at time of diagnosis. *Included International Classification of Diseases (10th revision) codes for the investigated first and second primary cancer sites are included in the appendix (p 1). †Diet high in red and processed meat. ‡Testing whether the exposure effect is unchanged over the four time bands after splitting the follow-up time using a Wald test. §This subanalysis is based on only female individuals with a hormone-related first primary cancer (n=91 823).

Table 3: Cause-specific hazard ratios with 95% CIs for development of a second primary cancer

did not have an increased hazard for second primary cancer despite the high 10-year cumulative incidence (appendix pp 7–8).

Increased HRs were seen for development of a second primary cancer with the same aetiology if the first primary cancer was alcohol-related (HR 1.09, 95% CI 1.06–1.13), smoking-related (1.73, 1.68–1.78), diet-related (1.32, 1.24–1.39), and virus-related (1.23, 1.13–1.35). Having had a hormone-related first cancer was associated with a lower HR of a second hormone-related cancer (0.77, 0.73–0.81; table 3). The cumulative probability of second primary cancers with the same aetiology was highest if the first primary cancer was alcohol-related, smoking-related, or hormone-related (appendix p 9).

Discussion

In this nationwide longitudinal population-based study we investigated the incidence of second primary cancer in Danish cancer survivors with up to 24 years of follow-up. The overall cumulative incidence of second primary cancer for all cancers combined was 6.3% at 5 years after diagnosis and 13.5% at 15 years after diagnosis. Out of all 27 investigated cancer types, survivors after cancers of the larynx, oropharynx and oral cavity, and bladder and urinary tract had the highest cumulative incidence of a second primary cancer 10 years after diagnosis, with cumulative incidence rates between 18.5% and 21.8%. Survivors who had had an alcohol-related, smoking-related, diet-related, or virus-related first primary cancer were at increased risk of developing

a second primary cancer with the same aetiology, whereas female individuals who had a hormone-related first cancer were at reduced risk of a second hormone-related cancer.

Strengths of the study include the large population-based cohort of cancer survivors and more than 56 000 cases of second primary cancer, taking into account the competing risk of death. Due to the comprehensive nature of the Danish administrative registries, we had reliable and unbiased information on both the first and the second primary cancers. The detailed procedures for notification of cancer cases in the Danish Cancer Registry minimises misclassification. We obtained precise linkage to the Danish population-based registries using the unique Danish PIN, enabling us to include the entire population of Danish cancer survivors, and we had almost complete follow-up. We were able to fully account for survivors' comorbidity up to 3 years before diagnosis through medically verified diagnoses from inpatient and outpatient contacts, and to adjust for important socioeconomic confounders.

Our study also had limitations. Due to the inherent nature of the information in the Danish administrative registries, we had no information on survivors' lifestyles such as smoking, exercise habits, or BMI. We used the cancer types as proxies for lifestyle and our results might therefore lack precision. Unmeasured common risk factors for second primary cancer and death could still induce some bias.²³ Furthermore, residual confounding might be present due to the categorisation of income in broad categories, as well as the scoring of the CCI being solely based on diagnosis codes but not including information on severity of the disease. We had no detailed information on stage at diagnosis and the primary cancer treatment received. There is firm evidence that a small but substantial proportion of new primary cancer cases can be attributed to cancer treatment;²⁴ some of the second primary cancers in our study are most likely a late effect of the cancer treatment provided, but it was not possible to investigate this hypothesis in detail. The purpose of this study was to investigate the risk of second primary cancer in cancer survivors overall and not the association between cancer treatment and new primary cancer.

In our study we used the Danish coding rules that apply to multiple cancers.¹⁴ Contrary to other definitions used by, for example, the SEER Programs,¹ there is no restricted time window to distinguish between two registrations of a new primary cancer. Furthermore, although the Danish coding rules are largely similar to the international rules for multiple primary cancers provided by IARC and the International Association of Cancer Registries,²⁵ there are important differences regarding grouping of different topography codes into single sites. One example is multiple colon cancers with different topography codes that are considered as one site. In the Danish coding rules, each new cancer

registration counts as a new cancer diagnosis provided that the topography codes are different and the otherwise prespecified coding rules are followed. Consequently, more primary tumours are registered in the Danish Cancer Registry and direct comparisons with other studies should therefore be done with caution. Other than the definition applied, the follow-up time, the patient population, and the disease stage at time of diagnosis also influence the number of new cancer registrations.¹ However, a strict and comprehensive evaluation of the validity of the Danish data is applied each time a new registration of new primary cancer occurs.¹⁴ Our findings for overall cumulative incidence of second primary cancer are similar to other large cohort studies. For example, Curtis and colleagues⁵ found slightly lower cumulative incidence of 5·0%, 8·4%, and 10·8% at 5, 10, and 15 years after diagnosis, respectively, compared with the 6·3%, 10·5%, and 13·5% reported in our study, potentially reflecting improvement in survival in most cancers since the 1970s. Like our study, they also reported increased risk of smoking-related and alcohol-related cancers in survivors of these same cancers.⁵ However, caution is advised when comparing absolute risk for new primary cancer across countries, as they will be influenced by population-level and cultural differences, such as lifestyle factors, including prevalence of smoking and obesity.

Many studies compare the incidence rate of new primary cancer in cancer survivors with the incidence rate in an age-sex-matched subgroup of the general population by calculating the ratio of observed to expected cases of new primary cancers, often reported as relative risk or standardised incidence ratio.^{12,26} In a previous study¹³ using the same cohort of cancer survivors as in the current study, we used survival analyses to investigate the somatic disease risk in Danish cancer survivors compared with an age-and-sex-matched cancer-free Danish general population, and found increased risk of almost all of the investigated somatic diseases. However, except for slightly increased estimated risks for survivors of colon cancer, bladder and urinary tract cancer, brain cancer, kidney cancer, and leukaemia, cancer survivors did not have an increased risk of second primary cancer (of any kind) compared with the general population. The long induction period of cancer and the shorter lifespan of many adult cancer survivors could partly explain these findings. However, the present study shows that specific groups of survivors are at increased risk of second primary cancer compared with others, which is valuable information for health-care personnel and survivors when planning for surveillance efforts and preventive care in follow-up programmes and beyond.

Our findings that cancer survivors with lifestyle-related and virus-related first cancers were at increased risk of developing a second cancer with the same aetiology corresponds to findings in other studies.^{5,7,27,28} Second

primary cancers in smoking-related first cancer sites is well known,^{7,29} often due to tissue susceptibility in the upper aerodigestive tract, and thus might reflect shared risk factors between the first and the second primary cancer³⁰—ie, smokers will continue to be at risk of smoking-related cancers. Shiels and colleagues²⁹ investigated the risk of smoking-related second cancers among survivors of lung, bladder, kidney, and head and neck cancers who smoked before their first cancer diagnosis, and found not only increased risk of new primary cancers but also that the risk decreased with increased number of years since smoking cessation. Other studies confirm these findings⁷ and highlight the importance of guidance and support to patients with cancer who wish to quit tobacco smoking, adding another potential benefit for the patient who can stop smoking.

We also found relatively high cumulative incidence of second primary cancer among patients with cancer types that are not smoking-related, such as malignant melanoma, prostate cancer, non-Hodgkin lymphoma, and Hodgkin lymphoma. Patients with these diagnoses have better survival statistics than most smoking-related cancer types, meaning that they might have long enough lifespans for them to develop a new cancer even though the risk of developing a second primary cancer is relatively low when compared with other cancer types. Conversely, the competing risk of death among patients who are diagnosed with cancers with poor survival, such as liver, pancreatic, and lung cancer, might obscure any risk pattern, because these patients often will not live long enough to develop a second cancer even though the first cancer might be lifestyle related.

Identifying risk patterns for new primary cancers in cancer survivors is of great individual importance for cancer survivors and their families as well as cancer clinicians who plan follow-up-care for patients, and is of broad public health relevance. Identifying cancer survivors who are at particular risk of developing a second primary cancer will enable prevention and screening efforts to be optimised and targeted, and contribute to timely personalised follow-up based on the individual cancer type and the treatment provided.

Contributors

TKK was the principal investigator; she contributed substantially to the study design, literature research, collection and assembly of data, and data analyses and interpretation, and wrote all drafts and the final version of the paper. EAWA did all the data management and statistical analysis, created all the tables and figures, and contributed to the conception and study design. GU, SBL, PEB, JFW, MB, CJ, and SOD contributed to the conception and study design, analysis, and interpretation of data. All authors contributed to the preparation of the manuscript and approved the final version. EAWA and TKK accessed and verified the raw data. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Declaration of interests

CJ received honoraria for lectures (Sanofi, Pfizer, Astellas, Janssen). TKK, EAWA, GU, SBL, PEB, JFW, MB, and SOD declare no competing interests.

Data sharing

The data used in this study were de-identified, linked, and accessed through the Danish Cancer Institute's secure server at Statistics Denmark. The authors do not have permission to share the data. In agreement with the General Data Protection Regulation, the study is registered in the Danish Cancer Society's internal project register database (journal number 2019-DCRC-0068).

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